MH4110 Partial Differential Equations

Handout 9 - Computation of Solutions

Emerging Paradigms in PDE Research

- **Integration with Machine Learning**
  - Physics-informed neural networks (PINNs) and operator learning
  - Data-driven discovery of governing equations
  - Overcoming the curse of dimensionality
- **High-dimensional PDEs**
  - Deep BSDE methods and neural solvers
  - Applications in finance, uncertainty quantification
- **Multiscale and multiphysics modeling**
  - Coupling PDEs across scales (micro $\rightarrow$ macro)
  - Hybrid numerical–data-driven frameworks

Advances in Analysis and Theory

- **Nonlinear PDEs and singularity formation**
  - Blow-up phenomena and long-time behavior
  - Regularity vs. singularity structures
- **Stochastic PDEs (SPDEs)**
  - Regularity structures and paracontrolled distributions
  - Connections with probability and quantum field theory
- **Geometric and variational methods**
  - Calculus of variations and optimal transport
  - PDEs on manifolds and geometric flows
- **Classical open problems**
  - Navier–Stokes regularity
  - Free boundary and nonlinear wave equations

Computational and Applied Frontiers

- **Next-generation numerical methods**
  - Adaptive, high-order, and structure-preserving schemes
  - Fast solvers for large-scale inverse problems
- **Inverse problems and imaging**
  - Adjoint-state methods and PDE-constrained optimization
  - Applications in geophysics, medical imaging, and materials
- **Nonlinear wave and soliton dynamics**
  - Soliton theory and dispersive PDEs
  - Applications in optics and quantum systems
- **Emerging directions**
  - Quantum computing for PDEs
  - Data assimilation and real-time forecasting

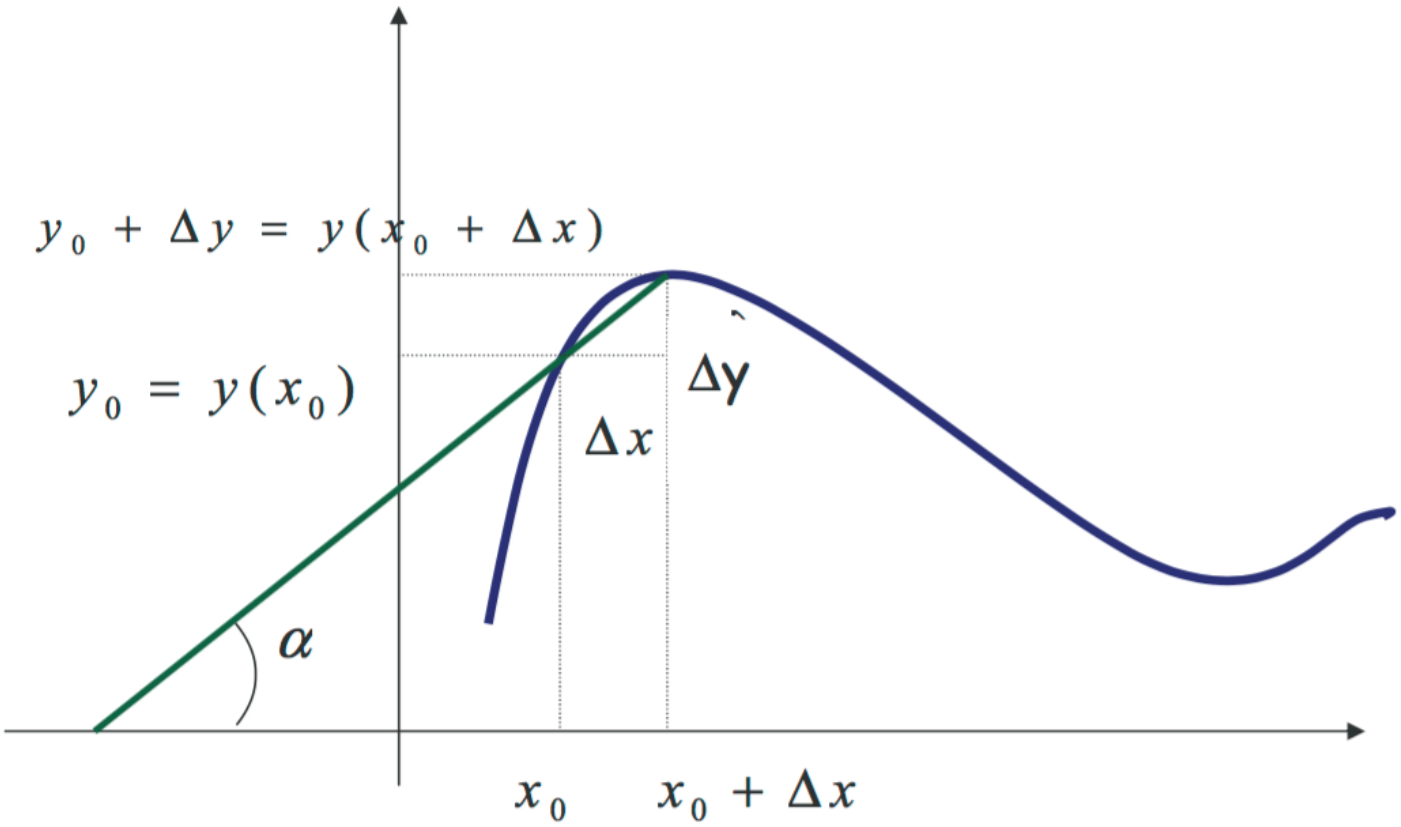
Outline

We have found formulas for many solutions to PDEs, but other problems encountered in practice are not as simple and cannot be solved by formula.

Reduce the process of solving a PDE with its auxiliary conditions to a finite number of arithmetical calculations that can be carried out by computer. All the problems we have studied can be so reduced.

- Finite differences
- Approximations of diffusions
- Approximations of waves

Derivative



$$\boxed{\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{\Delta x \to 0} \frac{y(x_0 + \Delta x) - y(x_0)}{\Delta x}}$$

§8.1 Finite differences

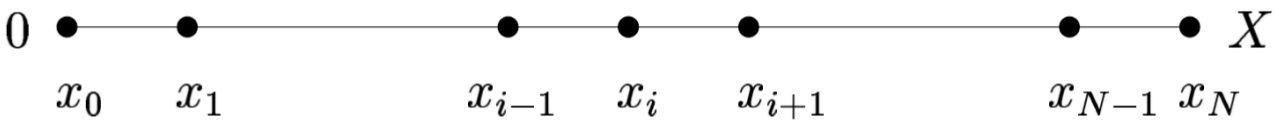
Finite differences: Replace each derivative by a difference quotient

Choose a mesh size Δx. Let's approximate the value $u(i\Delta x)$ for $x = x_i = i\Delta x$ by a number u_i indexed by an integer i:

$$u_i \approx u(i\Delta x).$$

Then the three standard approximations for the first derivative $\frac{\partial u(i\Delta x)}{\partial x}$ are:

- The backward difference: $\frac{u_i - u_{i-1}}{\Delta x}$
- The forward difference: $\frac{u_{i+1} - u_i}{\Delta x}$
- The centered difference: $\frac{u_{i+1} - u_{i-1}}{2\Delta x}$



§8.1 Finite differences

- The backward difference: $\frac{u_i - u_{i-1}}{\Delta x}$
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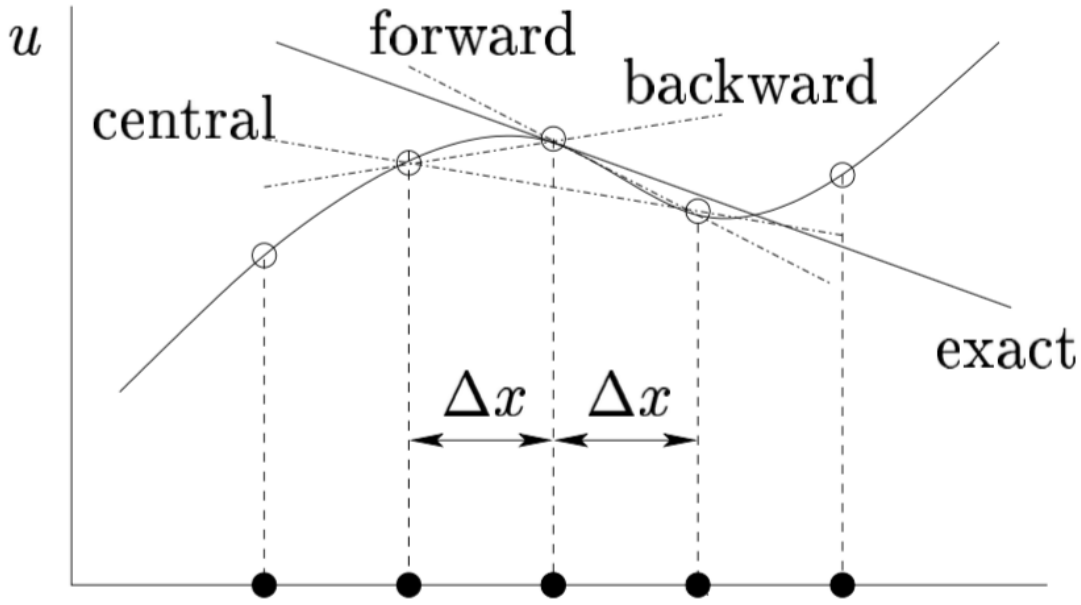
Geometric interpretation



§8.1 Finite differences

Finite differences: Replace each derivative by a difference quotient

The accuracy of the approximation can be investigated by using the Taylor expansion:

$$u(x+\Delta x) = u(x) + u'(x)\Delta x + \frac{1}{2}u''(x)(\Delta x)^2 + \frac{1}{6}u'''(x)(\Delta x)^3 + O(\Delta x)^4, \quad (1)$$

$$u(x-\Delta x) = u(x) - u'(x)\Delta x + \frac{1}{2}u''(x)(\Delta x)^2 - \frac{1}{6}u'''(x)(\Delta x)^3 + O(\Delta x)^4. \quad (2)$$

- The backward difference: $\frac{u_i - u_{i-1}}{\Delta x} \approx \frac{\partial u(i\Delta x)}{\partial x} = \frac{u(x_i) - u(x_i - \Delta x)}{\Delta x} + O(\Delta x)$
- The forward difference: $\frac{u_{i+1} - u_i}{\Delta x} \approx \frac{\partial u(i\Delta x)}{\partial x} = \frac{u(x_i + \Delta x) - u(x_i)}{\Delta x} + O(\Delta x)$
- The centered difference: $\frac{u_{i+1} - u_{i-1}}{2\Delta x} \approx \frac{\partial u(i\Delta x)}{\partial x} = \frac{u(x_i + \Delta x) - u(x_i - \Delta x)}{2\Delta x} + O(\Delta x)^2$

Finite differences for the second derivative

For the second derivative, the simplest approximation is the centered second difference:

$$u''(i\Delta x) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2}. \tag{3}$$

This is justified by the same two Taylor expansions (1)-(2) given above which, when added, give

$$u''(i\Delta x) \approx \frac{u(x_i + \Delta x) - 2u(x_i) + u(x_i - \Delta x)}{(\Delta x)^2} + O(\Delta x)^2. \tag{4}$$

That is, (4) is valid with an error of $O(\Delta x)^2$.

§8.1 Finite differences

For functions of two variables $u(x,t)$, we choose a mesh size for both variables. We write

$$u(i\Delta x, n\Delta t) \approx u_i^n, \tag{5}$$

where the n is a superscript, not a power. Then we can approximate, for instance,

$$\frac{\partial u}{\partial t}(i\Delta x, n\Delta t) \approx \frac{u_i^{n+1} - u_i^n}{\Delta t}, \tag{6}$$

the forward difference for $\partial u/\partial t$.

Similarly, the forward difference for $\partial u/\partial x$ is

$$\frac{\partial u}{\partial x}(i\Delta x, n\Delta t) \approx \frac{u_{i+1}^n - u_i^n}{\Delta x}.$$

Example 1

Let's consider the problem

$$\begin{aligned} &u_t = u_{xx}, \quad 0 < x < \pi,\ t > 0, \\ &u(0,t) = u(\pi,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x) = \begin{cases} x, & \text{in } (0, \frac{\pi}{2}), \\ \pi - x, & \text{in } (\frac{\pi}{2}, \pi). \end{cases} \end{aligned} \tag{7}$$

From Chapter 4, we know that the above problem has the series solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(- \Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin \frac{n\pi x}{l}, \tag{8}$$

provided that ϕ has the Fourier sine expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}. \tag{9}$$

§8.2 Approximations of diffusions

Since $k = 1$, $l = \pi$, the solution can be simplified as

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp(-n^2 t) \sin(nx). \tag{10}$$

After expanding ϕ into an Fourier sine series, the coefficients A_n can calculated as

$$A_n = \frac{2}{\pi} \int_0^{\pi} \phi(x) \sin(nx) dx = \begin{cases} \frac{4(-1)^{(n+1)/2}}{n^2 \pi}, & \text{if n is odd} \\ 0, & \text{if n is even.} \end{cases} \tag{11}$$

§8.2 Approximations of diffusions

We can also solve this problem using finite differences. We use a forward difference for u_t and a centered difference for u_{xx}.

Then the difference equation is

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2}. \tag{12}$$

It has a local truncation error of $O(\Delta t)$ (from the left side) and $O(\Delta x)^2$ (from the right side). Let

$$s = \frac{\Delta t}{(\Delta x)^2}, \tag{13}$$

we can solve (12) for u_i^{n+1} as

$$u_i^{n+1} = s(u_{i+1}^n + u_{i-1}^n) + (1 - 2s)u_i^n. \tag{14}$$

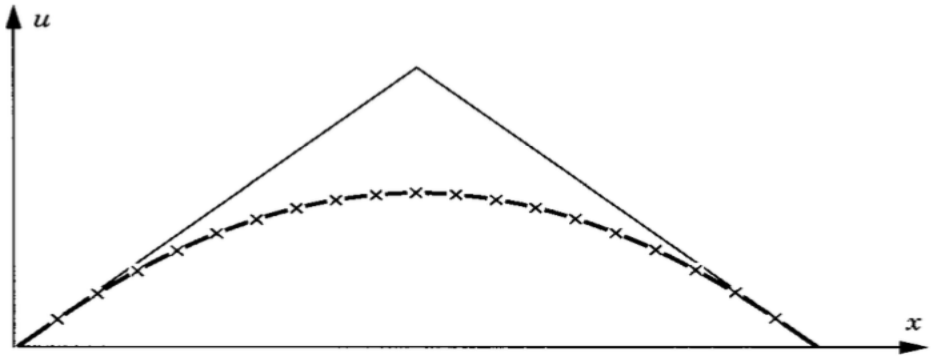
The scheme is said to be explicit because the values at the $(n+1)$st time step are given explicitly in terms of the values at the earlier times.

§8.2 Approximations of diffusions

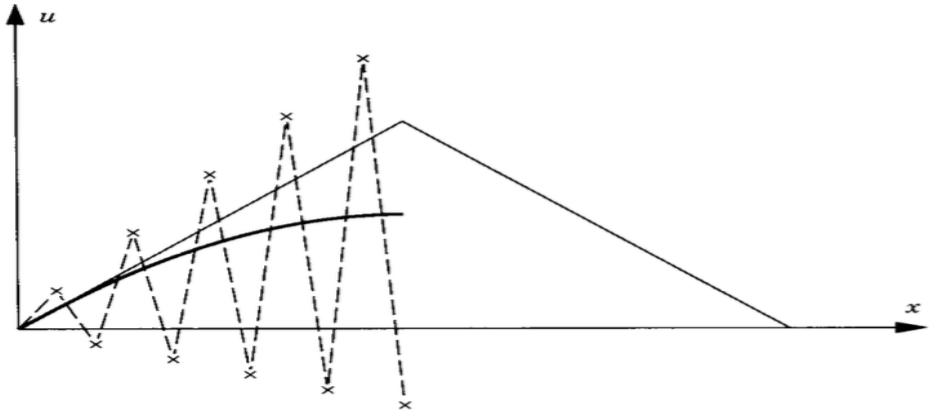
We approximate this problem by the scheme (14) for $i = 1, ..., I - 1$ and $n = 0, 1, 2, ...$ together with the discrete boundary and initial conditions

$$u_0^n = u_I^n = 0 \quad \text{and} \quad u_i^0 = \phi(i\Delta x).$$

- For $I = 20$, $\Delta x = \pi/20$, and $s = 5/11$ (exactly on target, stable)



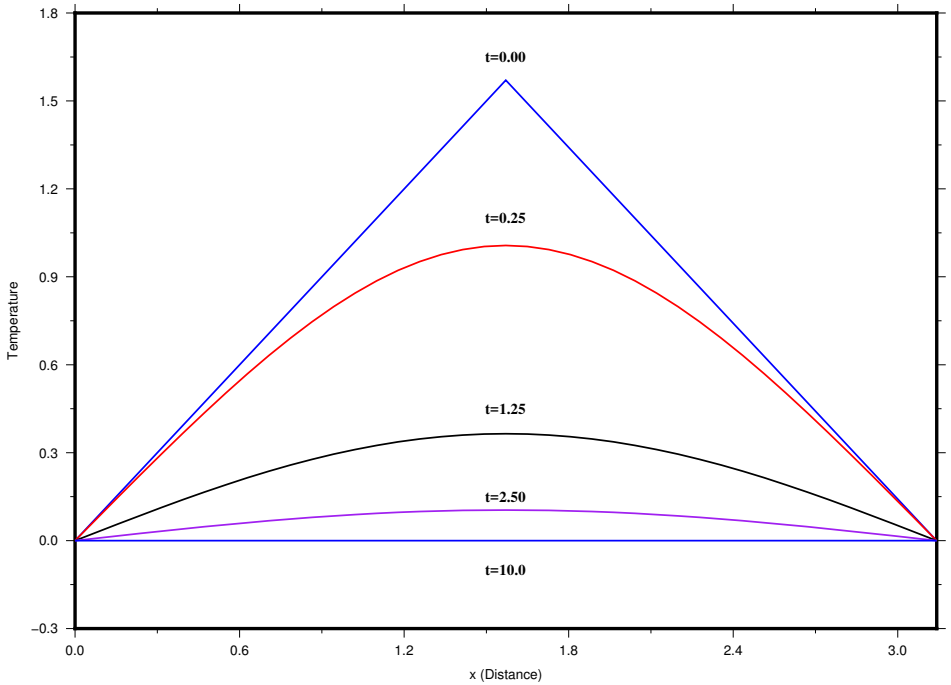
- For $I = 20$, $\Delta x = \pi/20$, and $s = 5/9$ (wild oscillations, unstable)



§8.2 Approximations of diffusions

The Dirichlet problem for the heat equation

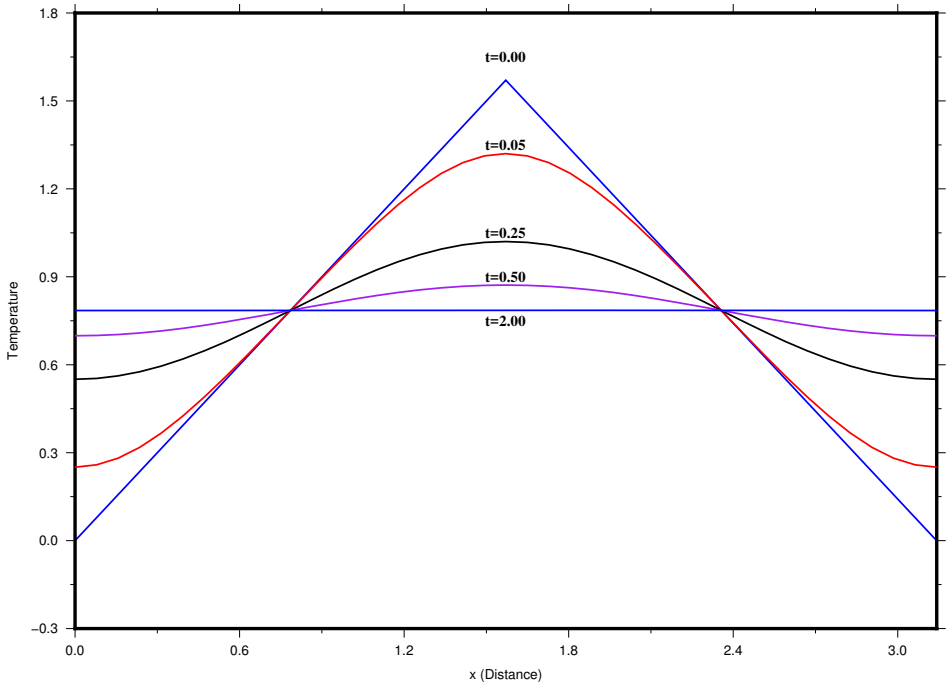
$$
\begin{aligned}
&u_t = u_{xx}, \quad 0 < x < \pi, \ t > 0, \\
&u(0,t) = u(\pi,t) = 0, \quad t > 0, \\
&u(x,0) = \phi(x) = \begin{cases} x, & \text{in } (0, \frac{\pi}{2}), \\ \pi - x, & \text{in } (\frac{\pi}{2}, \pi). \end{cases}
\end{aligned}
\tag{15}
$$



§8.2 Approximations of diffusions

The Neumann problem for the heat equation

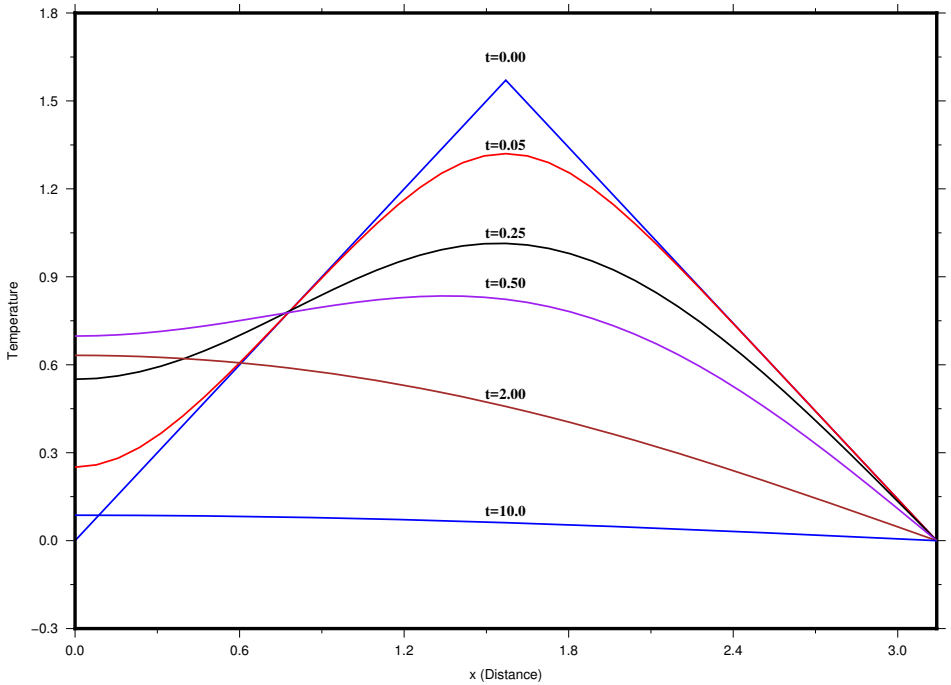
$$
\begin{aligned}
&u_t = u_{xx}, \quad 0 < x < \pi,\ t > 0,\\
&u_x(0,t) = u_x(\pi,t) = 0, \quad t > 0,\\
&u(x,0) = \phi(x) = \begin{cases} x, & \text{in } (0, \frac{\pi}{2}),\\ \pi - x, & \text{in } (\frac{\pi}{2}, \pi). \end{cases}
\end{aligned}
\tag{16}
$$



§8.2 Approximations of diffusions

The mixed boundary problem for the heat equation

$$
\begin{aligned}
&u_t = u_{xx}, \quad 0 < x < \pi,\ t > 0,\\
&u_x(0,t) = u(\pi,t) = 0, \quad t > 0,\\
&u(x,0) = \phi(x) = \begin{cases} x, & \text{in } (0, \frac{\pi}{2}),\\ \pi - x, & \text{in } (\frac{\pi}{2}, \pi). \end{cases}
\end{aligned}
\tag{17}
$$



Initial boundary value problem for the wave equation

Let's consider the problem

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\ &u(0,t) = g(t), \quad u(\pi,t) = h(t), \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x). \end{aligned} \tag{18}$$

The simplest scheme is the one using centered differences for both u_{tt} and u_{xx}:

$$\frac{u_i^{n+1} - 2u_i^n + u_i^{n-1}}{(\Delta t)^2} = c^2 \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2}. \tag{19}$$

It is explicit since the $(n+1)$st time step appears only on the left side.

§8.3 Approximations of waves

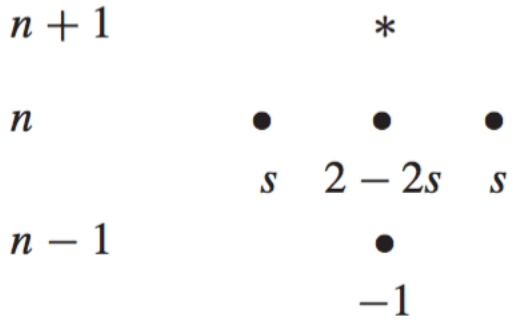
The simplest scheme is the one using centered differences:

$$\frac{u_i^{n+1} - 2u_i^n + u_i^{n-1}}{(\Delta t)^2} = c^2 \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2}. \tag{20}$$

If we define $s = c^2(\Delta t)^2/(\Delta x)^2$, then

$$u_i^{n+1} = s(u_{i+1}^n + u_{i-1}^n) + 2(1-s)u_i^n - u_i^{n-1}. \tag{21}$$

The template diagram is



Notice that the value at the $(n+1)$st time step depends on the two previous steps, because the wave equation has time derivatives of second order. Therefore, the first two rows u_i^0 and u_i^1 must be given as initial conditions.

Initial conditions

We approximate the conditions $u(x,0) = \phi(x)$ and $\partial u/\partial t(x,0) = \psi(x)$ by

$$u_i^0 = \phi(i\Delta x), \quad \frac{u_i^1 - u_i^{-1}}{2\Delta t} = \psi(i\Delta x). \tag{22}$$

This approximation is chosen to have a $O(\Delta x)^2$ local truncation error in order to match the $O(\Delta x)^2 + O(\Delta t)^2$ truncation error of the scheme (21).

Let's abbreviate $\phi_i = \phi(i\Delta x)$ and $\psi_j = \psi(j\Delta x)$. Now (21) in the case $n = 0$ is

$$u_i^1 + u_i^{-1} = s(u_{i+1}^0 + u_{i-1}^0) + 2(1-s)u_i^0. \tag{23}$$

Together with (22), this gives us the starting values

$$u_i^0 = \phi_i, \quad u_i^1 = \frac{s}{2}(\phi_{i+1} + \phi_{i-1}) + (1-s)\phi_i + \psi_i \Delta t, \tag{24}$$

the first two rows of the computation. Then we march ahead in time to get u_i^2, u_i^3 , and so on, using (21).

The Dirichlet problem for the wave equation

Solve the problems

$$\begin{aligned} &u_{tt} = u_{xx}, \quad -5 < x < 5,\ t > 0, \\ &u(x,0) = \phi(x) = \exp(-4x^2), \quad u_t(x,0) = \psi(x) = 0, \quad -5 < x < 5. \end{aligned} \tag{25}$$

- The Dirichlet case $u(-5,t) = u(5,t) = 0, \quad t > 0$
- The Neumann case $u_x(-5,t) = u_x(5,t) = 0, \quad t > 0$
- The mixed boundary $u_x(-5,t) = u(5,t) = 0, \quad t > 0$

The above three problems can be solved by the method of separation of variables. But it is not so easy (sometimes even impossible) to explicitly calculate the coefficients of the Fourier series.

Contrarily, it is not difficult to find the solutions numerically. Numerical solutions are shown in animations in the class.